

Alexander Tchourbanov

Hands-On Controls | Electrical Installation | System Troubleshooting

Brownsville, TX • (520) 820-7281 • achurbanov@gmail.com

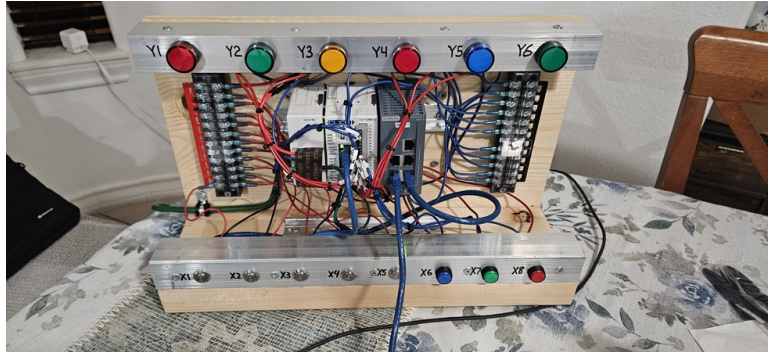
This portfolio documents selected projects that I personally assembled, wired, installed, configured, tested, and troubleshot. It complements my formal training in Automation and Remote Control and demonstrates practical work with real hardware, tools, electrical systems, and mechanical installation.

PLC Controls Trainer — Programming, Wiring, and Commissioning

I built and commissioned this bench-scale controls demonstrator around an AutomationDirect CLICK PLC. The trainer provides a physical platform for writing, testing, and troubleshooting machine-control logic rather than relying only on simulation.

- Mounted the PLC, terminal blocks, operator push buttons, and pilot lights.
- Completed point-to-point field wiring and organized input/output connections.
- Configured PLC-to-workstation Ethernet addressing and verified communications.
- Mapped and tested discrete I/O on live hardware.
- Programmed and debugged ladder logic for sequencing, interlocks, alarms, timers, counters, latching, reset functions, and lead/lag operation.
- Used online monitoring and systematic checks to verify commanded and actual states.

Hands-on evidence: panel layout, terminations, I/O checkout, communications setup, functional testing, and PLC troubleshooting.



Operational CLICK PLC trainer with labeled field inputs and outputs, terminal blocks, Ethernet connection, push buttons, and pilot lights.

Relevant Practical Capabilities

Controls	PLC ladder logic, discrete I/O, sequencing, permissives/interlocks, alarms, communications, online monitoring, and functional verification.
Electrical	Schematics, terminations, multimeter-based diagnosis, grounding, disconnects, low-voltage controls, safe isolation, and continuity checks.
Mechanical	Measuring, drilling, fastening, mounting, trenching, equipment installation, plumbing, weather-proof enclosures, and hand/power tools.
Work methods	Plan, build, test, diagnose, correct, document, and verify complete systems using a methodical, safety-conscious approach.

Electrical Power System — Installation and Functional Verification

I installed and tested a residential photovoltaic system incorporating a 20-panel array, underground electrical-feed work, grounding electrodes, combiner/disconnect equipment, a 6 kW split-phase inverter, and an approximately 15 kWh 48 V battery bank. The project required planning component placement, installing mechanical and electrical hardware, routing conductors, making connections, checking polarity and continuity, and verifying operation as an integrated system.



Installed and wired PV disconnect switches and combiner equipment.



Mounted and connected the 6 kW split-phase inverter.



Integrated the approximately 15 kWh, 48 V battery bank.



Trenched the route for underground electrical cable and conduit.



Installed grounding rods as part of the system grounding work.



Installed the 20-panel, approximately 4 kW photovoltaic array.

System Completion, Testing, and Production



Commissioning result: system operating and producing approximately 2.89 kW.

What This Project Demonstrates

- Direct involvement from physical installation through energized functional testing.
- Integration of electrical, mechanical, grounding, protection, and energy-storage components.
- Careful use of tools, inspection, measurement, and systematic verification before operation.

Low-Voltage Pump Controls, Fabrication, and Field Installation

I designed and installed an approximately 3,000-gallon rainwater collection and irrigation system with a solar-powered 12 V pump. The work combined low-voltage electrical equipment, a weather-resistant enclosure, plumbing, filtration, overflow management, structural mounting, and sheet-metal fabrication. I tested the completed system under operating conditions and corrected installation issues as they appeared.



Installed the 12 V battery, pump controls, protection, and wiring in a field enclosure.



Fabricated and installed the solar-powered pump mast and equipment mounting.



Installed plumbing and overflow catchment hardware.



Measured, cut, riveted, and sealed the roof collection opening.

System Completion, Testing, and Communications



Transported 330-gallon storage totes.



Installed gutters routing water across the house.



Installed and connected the main water-storage tanks.



Commissioned rainwater collection during rainfall.



Built and serviced the dual debris-filtration system.

Independent Low-Voltage System Integration

I also installed a self-powered wireless security camera with cellular connectivity. The work included selecting a location, mounting equipment, routing and securing wiring, accounting for weather exposure, configuring the communications connection, and confirming dependable operation.

Hands-on evidence: mechanical mounting, low-voltage power, communications setup, cable routing, environmental protection, and end-to-end testing.



Solar-powered cellular security camera installed and operating.

I am comfortable working directly with hardware: assembling, wiring, installing, testing, troubleshooting, and improving complete systems.